THE WEIGHTING OF A RELATIVE ZERO-CYCLE IS COMPATIBLE WITH BASE CHANGE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The separable-first subfield of a finitely generated field extension (Fields, Lemma fields-lemma-separable-first, canonical tag 030K). cited at proof lines 1709-1710. Exhibits k tensor kappa(x) as a product of local factors each of degree equal to the inseparable degree, proving the key length equality.
- Pullback formula for lengths along flat ring maps (Algebra, Lemma algebra-lemma-pullback-module, canonical tag 02M1). cited at proof lines 1706-1707. Transports the identity from the algebraic closure base change back to the original tensor product, using flatness of the map between tensor products.

Source reference. Original lemma 030K: sources/stacks-chow/tags.txt, source lines 3951–3951. Cited at proof lines 1709-1710: decomposes the algebraic-closure tensor product into local factors of inseparable degree.

Source reference. Original lemma 02M1: sources/stacks-chow/tags.txt, source lines 3428–3428. Cited at proof lines 1706-1707: transports the identity back from the algebraic closure base change.

Lemma 0.2. *In the situation above, if $g : S' \to S$ is a morphism of locally Noetherian schemes, then $w_{g^*\alpha} = w_\alpha \circ g'$ where $g' : X' \to X$ is the projection $X' = S' \times_S X \to X$.*

Let S be a locally Noetherian scheme. Let $f : X \to S$ be a locally quasi-finite morphism of schemes. Then we have $z(X/S, 0) = z_{equi}(X/S, 0)$ and $z(X/S, r) = 0$ for $r > 0$. Given $\alpha \in z(X/S, 0)$ let us define a map

$$w_\alpha : X \longrightarrow \mathbf{Z}, \quad x \mapsto \alpha(x)[\kappa(x) : \kappa(s)]_i \quad \text{where } s = f(x)$$

Here $\alpha(x)$ denotes the coefficient of x in the 0-cycle α_s on the fibre X_s and $[K : k]_i$ denotes the inseparable degree of a finite field extension.

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Proof. Let $x' \in X'$ with images s', s, x in S', S, X. Then the coefficient of $[x']$ in the base change of $[x]$ by $\kappa(s')/\kappa(s)$ is the length of the local ring $(\kappa(s') \otimes_{\kappa(s)} \kappa(x))_{\mathfrak{q}}$. Here $\mathfrak{q}$ is the prime ideal corresponding to x'. Thus compatibility with base change follows if

$$[\kappa(x) : \kappa(s)]_i = \text{length}((\kappa(s') \otimes_{\kappa(s)} \kappa(x))_{\mathfrak{q}})[\kappa(x') : \kappa(s')]_i$$

Let $k/\kappa(s')$ be an algebraically closure. Choose a prime $\mathfrak{p} \subset k \otimes_{\kappa(s)} \kappa(x)$ lying over $\mathfrak{q}$. Suppose we can show that

$$[\kappa(x) : \kappa(s)]_i = \text{length}((k \otimes_{\kappa(s)} \kappa(x))_{\mathfrak{p}}) \quad \text{and} \quad [\kappa(x') : \kappa(s')]_i = \text{length}((k \otimes_{\kappa(s')} \kappa(x'))_{\mathfrak{p}})$$

Then we win because

$$\text{length}((\kappa(s') \otimes_{\kappa(s)} \kappa(x))_{\mathfrak{q}})\text{length}((k \otimes_{\kappa(s')} \kappa(x'))_{\mathfrak{p}}) = \text{length}((k \otimes_{\kappa(s)} \kappa(x))_{\mathfrak{p}})$$

by Algebra, Lemma 02M1 and flatness of $\kappa(s') \otimes_{\kappa(s)} \kappa(x) \to k \otimes_{\kappa(s)} \kappa(x)$. To show the two equalities, it suffices to prove the first. Let $\kappa(x)/\kappa/\kappa(s)$ be the subfield constructed in Fields, Lemma 030K. Then we see that

$$k \otimes_{\kappa(s)} \kappa(x) = \prod\nolimits_{\sigma : \kappa \to k} k \otimes_{\sigma, \kappa} \kappa(x)$$

and each of the factors is local of degree $[\kappa(x) : \kappa] = [\kappa(x) : \kappa(s)]_i$ as desired. $\square$